

Installation note for OceanWave3D-seakeeping¹

Mostafa Amini Afshar and Harry B. Bingham²

Section of Fluid Mechanics, Coastal and Maritime Engineering³

Department of Mechanical Engineering⁴

Technical University of Denmark⁵

February 1, 2017⁶

1 Introduction⁷

In this document the required process for installation of the seakeeping solver is described.⁸ Please note that OceanWave3D-Seakeeping code is making use of several external libraries. These libraries must be built and installed before the seakeeping code is compiled and used. The main library is called **Overture** (Brown et al., 1999). It is assumed here that the Overture library has already been installed. For Overture installation procedure, please refer to the relevant document in this folder. It is also recommended to install the **PETSc** library as it has been explained in the Overture installation guide. The seakeeping code is also using the GNU Scientific Library (gsl) (Gough, 2009) whose installation will be described in this note. For all other required libraries please refer to the documentation for the Overture installation in this folder.

2 Building the gsl library⁹

This library is a collection of routines for the numerical calculation in C and C++. In the seakeeping code it is used for instance to perform the fft, least-square fitting, complex number operations and so on. The gsl library can be downloaded from <https://ftp.gnu.org/gnu/gsl/>.¹⁰ Note that the seakeeping code is written based on gsl version 1.16. For installation:

```
tar xzf gsl-1.16.tar.gz11
```

```
cd gsl-1.1612
```

```
./configure13
```

```
make14
```

```
make check15
```

```
make install16
```

For more information regarding the installation process for gsl refer to the INSTALL file in the gsl library folder.¹⁷

3 Building the seakeeping code¹⁸

The whole seakeeping code is arranged in the following folders:¹⁹

cpp¹

This folder contains the whole implementation files of the code.²

include³

Is the place for the header files. All implementation files in cpp folder are declared in the files⁴ located in include folder.

bin⁵

All the binary and object files resulting from the compilation will be located in this folder. Note⁶ that two executables **RUN.exe** and **POST.exe** are built in the parent folder.

The compilation of the code is simple and can be done just by typing:⁷

make⁸

in the parent folder. If successful, this would make the two executables **RUN.exe** and **POST.exe**⁹ in the parent folder. Finally you can also add the paths of these executables to your existing path, by adding the following line in to the .bashrc file in the home directory:

```
export=PATH:[the path to the executables (RUN.exe and POST.exe)].10
```

References¹

Brown, D. L., W. D. Henshaw, and D. J. Quinlan. Overture: An object-oriented framework for solving partial differential equations on overlapping grids. *Object Oriented Methods for Interoperable Scientific and Engineering Computing, SIAM*, pages 245–255, 1999.²

Gough, B. *GNU scientific library reference manual*. Network Theory Ltd., 2009.